

Claude to Instagram: AI-Powered Content Generation

A zero-touch content pipeline from prompt to post in 12 seconds

\$27/mo

Total cost

12s

Prompt to post

3×/day

Auto-posts

0

Manual steps

The Problem

Creating consistent social content is slow, expensive, and repetitive



4+ hours/week

Writing captions, choosing images, scheduling posts manually



High agency cost

Outsourcing content creation runs \$500–\$2,000/month



Inconsistent output

Manual process leads to missed posts and off-brand content



No scalability

More posts = more time. Linear cost that never improves

The Solution

A fully automated pipeline: Claude generates content, DALL-E creates images, Meta publishes



Result: Fresh AI-generated post with unique image, caption, and hashtags live on Instagram — automatically, 3× per day, every day.

Schedule

Claude

JSON Parse

DALL-E

Cloudinary

Instagram

Node 1 — Schedule Trigger

Fires the workflow automatically every 8 hours — no manual input needed

n8n Configuration

Node type

Schedule Trigger

Trigger interval

Hours

Hours between triggers

8

Posts per day

3

Connected to

HTTP Request (Claude)

What it does

Replaces the manual 'Execute workflow' button. Once the workflow is Published in n8n, this node fires automatically every 8 hours — 3 posts per day without touching anything.

⚠ Critical step

The workflow must be Published (top-right in n8n) for the schedule to run. An unpublished workflow only runs when you click Execute manually.

✓ Schedule

Claude

JSON Parse

DALL-E

Cloudinary

Instagram

Node 2 — Claude Haiku API

Generates hook, caption, hashtags and image prompt as a single JSON object

HTTP Request config

Method: **POST**

URL: **api.anthropic.com/v1/messages**

Model: **claude-haiku-4-5-20251001**

Required headers

x-api-key YOUR_ANTHROPIC_KEY

anthropic-version 2023-06-01

content-type application/json

System prompt (key extract)

"Rotate between: stat-based, story-based, tip-based, myth-busting, question-based. Return ONLY raw JSON. No markdown. Start with { end with }."

Output fields

hook · script · caption · hashtags · visual_desc

✓ Schedule

✓ Claude

JSON Parse

DALL-E

Cloudinary

Instagram

Node 3 — JSON Parse

Extracts clean fields from Claude's response — handles markdown and array edge cases

Code node — JavaScript

```
let raw = $input.first().json.content[0].text;

// Strip markdown backticks if Claude adds them
raw = raw.replace(/```json/g, '')
          .replace(/```/g, '').trim();

const parsed = JSON.parse(raw);

// Join hashtags if Claude returns an array
if (Array.isArray(parsed.hashtags)) {
  parsed.hashtags = parsed.hashtags.join(' ');
}

return [{ json: parsed }];
```

Why strip backticks?

Claude sometimes wraps JSON in ```json blocks despite instructions. The replace() is a safety net so parsing never fails.

Why fix hashtags?

Claude sometimes returns hashtags as a JSON array instead of a string. The Array.isArray() check joins them into a single string Meta accepts.

✓ Schedule

✓ Claude

✓ JSON Parse

DALL-E

Cloudinary

Instagram

Node 4 — gpt-image-2 (Image Gen)

Generates a unique 1024×1024 JPEG from Claude's visual_desc field

HTTP Request config

Method:	POST
URL:	api.openai.com/v1/images/generations
Header: Authorization:	Bearer YOUR_OPENAI_KEY

Cost at scale

1 post/day (30 images)	~\$0.60/mo
3 posts/day (90 images)	~\$1.80/mo
10 posts/day (300 images)	~\$6.00/mo

JSON body

```
{
  "model": "gpt-image-2",
  "prompt": "{{ $json.visual_desc }}...",
  "size": "1024x1024",
  "quality": "low",
  "output_format": "jpeg",
  "n": 1
}
```

⚠ Key gotchas

- gpt-image-2 always returns base64, never a URL
- quality: 'low' is 15× faster and cheaper than 'high'
- OpenAI org verification required before first call
- Output goes to next node as base64 string

✓ Schedule

✓ Claude

✓ JSON Parse

✓ DALL-E

Cloudinary

Instagram

Node 5 — Cloudinary Upload

Converts base64 image to a public JPEG URL that Meta's API accepts

Code node — JavaScript

```
const base64Data =
  $input.first().json.data[0].b64_json;

const cloudName = 'YOUR_CLOUD_NAME';
const uploadPreset = 'ml_default';

const response = await this.helpers.httpRequest({
  method: 'POST',
  url: `https://api.cloudinary.com/v1_1/
    ${cloudName}/image/upload`,
  body: {
    file: `data:image/jpeg;base64,${base64Data}`,
    upload_preset: uploadPreset
  },
  json: true
});

return [{ json: { image_url: response.secure_url } }];
```

Why not Google Drive?

Meta's API blocks Drive URLs (redirects).
Cloudinary serves direct URLs with no
redirects. Free tier: 25GB storage.

Setup required

1. Sign up at cloudinary.com (free)
2. Settings → Upload → Upload presets
3. Set ml_default to Unsigned
4. Copy your Cloud Name into the code

✓ Schedule

✓ Claude

✓ JSON Parse

✓ DALL-E

✓ Cloudinary

Instagram

Node 6 — Instagram Publish

Two-step Meta Graph API call: create container → publish. Both required.

Step A — Create Media Container

```
POST /{ig-user-id}/media
```

```
{
  "image_url": "{{ $json.image_url }}",
  "caption": "{{ caption }}\n\n{{ hashtags }}",
  "access_token": "YOUR_TOKEN"
}
```

```
Returns: { id: 'container_id' }
```

Step B — Publish Container

```
POST /{ig-user-id}/media_publish
```

```
{
  "creation_id": "{{ $json.id }}",
  "access_token": "YOUR_TOKEN"
}
```

```
Returns: { id: 'published_media_id' }
```

⚠ Key requirements: Instagram Business account + Facebook Page linked + System User token (Never expire) + image must be JPEG not PNG

Architecture

Built on n8n Cloud — 5 HTTP nodes, no custom infrastructure

Node	Service	What it does	Key config
1. Trigger	n8n Schedule	Fires every 8 hours	Interval: 8hrs
2. Content	Claude Haiku API	Generates hook, caption, hashtags, image prompt	max_tokens: 600
3. Parse	JavaScript Code	Strips markdown, joins hashtag array	JSON.parse + array fix
4. Image	gpt-image-2 API	Generates 1024×1024 JPEG from prompt	quality: low
5. Host	Cloudinary API	Uploads base64 → returns public URL	unsigned preset
6. Post	Meta Graph API	Creates container + publishes to Instagram	Two-step publish

Live Results

@eve_rydayai — built and live in one session

AI FOR ANALYTICS

Stop spending 8 hours on reports.

Companies using AI analytics cut reporting time by 75%...

Post 1

AI FOR ANALYTICS

Their reports take 4 hours. Ours take 4 minutes.

Teams using AI analytics cut reporting time by 80%...

Post 2

AI FOR ANALYTICS

Your dashboard isn't slow. You're drowning in data.

AI analytics centralizes your data...

Post 3

Cost Breakdown

Full automation at a fraction of traditional agency cost

n8n Cloud	Workflow automation	\$24.00
Claude Haiku	Content generation	\$1.50
gpt-image-2	Image generation (90 images)	\$1.80
Cloudinary	Image hosting	Free
Meta Graph API	Instagram publishing	Free
Total monthly cost		~\$27.30/mo

Scale Potential

Same pipeline, multiply outputs — cost grows linearly, value grows exponentially

Starter

1/day

30 posts

\$25/mo

CURRENT

Growth

3/day

90 posts

\$27/mo

Scale

10/day

300 posts

\$35/mo

Going from 1 to 10 posts/day costs only \$10 more per month — the infrastructure cost is nearly fixed.

Monetization Roadmap

Three revenue streams, each building on the last

Month 1–2

Affiliate Marketing

Promote AI tools in captions. Claude, ChatGPT, analytics platforms all have programs. \$5–50 per conversion.

\$200–500/mo

Month 2–3

Digital Products

Sell prompt libraries and templates on Gumroad. '100 AI Analytics Prompts' at \$29–99. Audience is already warm.

\$500–2K/mo

Month 4+

Consulting & Setup

Offer done-for-you pipeline setup. Businesses pay \$500–2K to have this built for them. 3 clients covers all costs.

\$2K–10K/mo

Next Steps

What comes after the MVP

01

Auto-DM on comments

When someone comments 'PROMPT', Meta Webhooks trigger Claude to DM the template automatically. Zero manual follow-up.

Next sprint

02

Carousel posts

3–5 slides per post using multiple DALL-E calls. Carousels get 3× more saves than single images.

Week 2

03

Reels with Veo

Google Veo API generates 8s video from visual_desc. Reels get 5× organic reach vs static posts.

Month 2

04

Multi-account scale

Same pipeline, different Claude system prompts. Run 5–10 niche accounts from one n8n workflow.

Month 3

From zero to automated

Claude to Instagram

AI-Powered Content Generation

\$27/mo

All-in cost

90

Posts/month

0

Manual steps

12s

Per post

@eve_rydayai